Module 6 : Generative AI

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Agenda - Part : III

- Introduction to Agentic AI
- Distinction between traditional AI, LLMs, and agentic AI
- Role of NLP in enabling agentic AI
- Use cases: conversational agents, task-oriented agents, and collaborative agents in domains like hospitality, education, and customer service.

What is Agentic AI?

- A system where an AI model (the "brain") is given a high-level **goal** and the **autonomy** to plan, use tools, and execute actions to reach it.
- **The "Reasoning Loop":** Unlike a single prompt-response, Agentic AI operates in a loop:
    1. **Plan:** Break the goal into sub-tasks.
    2. **Execute:** Call a tool (API, search, code).
    3. **Observe:** Check the result of the tool.
    4. **Refine:** If the result is wrong, try a different approach.

Traditional AI vs. LLMs vs. Agentic AI

Feature	Traditional AI	Large Language Models (LLM)	Agentic AI
Primary Logic	Rule-based (If-Then)	Statistical (Probabilistic)	Goal-driven (Autonomous)
Interaction	Fixed Input -> Output	Prompt ->Response	Goal ->Multi-step Plan
Autonomy	Zero (Deterministic)	Low (Needs human prompt)	High (Operates in loops)
Capabilities	Data patterns/Math	Text/Code generation	Task execution/Tool use
Example	Fraud detection	ChatGPT (2022-2023)	CrewAI/AutoGen (2025+)

Role of NLP in Enabling Agentic AI

- **The Interface of Intent:** Natural Language Processing (NLP) is the bridge that allows a human to say, *"Organize my travel,"* and the agent to understand the **intent** without code.
- **Semantic Reasoning:** LLMs use NLP to "reason" about which tool to use next based on the text context.
- **Syntactic Translation:** The agent uses NLP to translate human instructions into structured parameters for APIs (e.g., turning "next Tuesday" into 2026-04-21).
- **Self-Correction:** Agents use NLP to "talk to themselves" through internal logs to evaluate if their previous action was successful.

Category 1: Conversational Agents

- **Focus:** Deep dialogue and relationship management.
- **Capability:** They don't just answer; they maintain long-term memory of a user's preferences and emotional state.
- **Key Feature: Multimodal Interaction**—switching between voice, text, and even vision seamlessly.
- **Example:** An AI companion that helps a user manage their mental health through daily check-ins.

Category 2: Task-Oriented Agents

- **Focus:** Efficiency and "Getting it done."
- **Capability:** These agents are specialized to interact with software stacks (CRM, ERP, Billing).
- **Workflow:**
    1. User: *"Refund my last order."*
    2. Agent: Queries database -> Validates policy -> Executes Stripe API -> Updates ZenDesk ticket.
- **Example:** A "Browser Agent" that can literally click through a website to book a flight for you.

Category 3: Collaborative Agents (MAS)

- **Focus:** Teamwork (Multi-Agent Systems).
- **Capability:** Multiple agents with different "personalities" or roles talk to each other to solve a complex problem.
- **Roles:** A "Manager" agent coordinates a "Researcher" agent and a "Writer" agent.
- **Strength:** Higher accuracy because agents can peer-review each other’s work before showing it to the human

Use Case: Hospitality (The 2026 Concierge)

- **Hyper-Personalization:** Agents analyze loyalty data and social preferences to proactively offer a room upgrade *before* the guest asks.
- **Disruption Management:** If a flight is delayed, the agent automatically:
  1. Extends the hotel checkout.
  2. Rebooks the airport transfer.
  3. Updates the dinner reservation.
- **Direct Revenue:** Upselling services (spa, tours) through conversational messaging.

Use Case: Higher Education (The Socratic Tutor)

- **Socratic Tutors:** Instead of giving the answer to a math problem, the agent asks guiding questions to help the student find it.
- **Predictive Intervention:** An agent monitors a student's engagement in the Learning Management System (LMS) and reaches out to offer help *before* they fail a midterm.
- **Administrative Ops:** Automating credit transfer evaluations and scholarship matching for thousands of applicants in seconds.

Use Case: Customer Service (The Unified Front)

- **Beyond the Chatbot:** In 2026, customer service agents are "Action-Bots."
- **Capability:** They resolve 80% of tickets end-to-end (changing addresses, upgrading plans, troubleshooting hardware via camera feed).
- **Impact:** Reduced wait times and 24/7 "expert-level" support in 100+ languages.
- **The Human Role:** Humans now only handle "Edge Cases"—complex, emotionally sensitive issues where empathy is the primary tool.

The Path Forward

- **Agentic AI = LLM + Planning + Tools + Memory.**
- **NLP is the "OS"** that runs these agents.
- The biggest challenge for 2026 is **Reliability** and **Safety**—ensuring agents don't make unauthorized purchases or share private data